

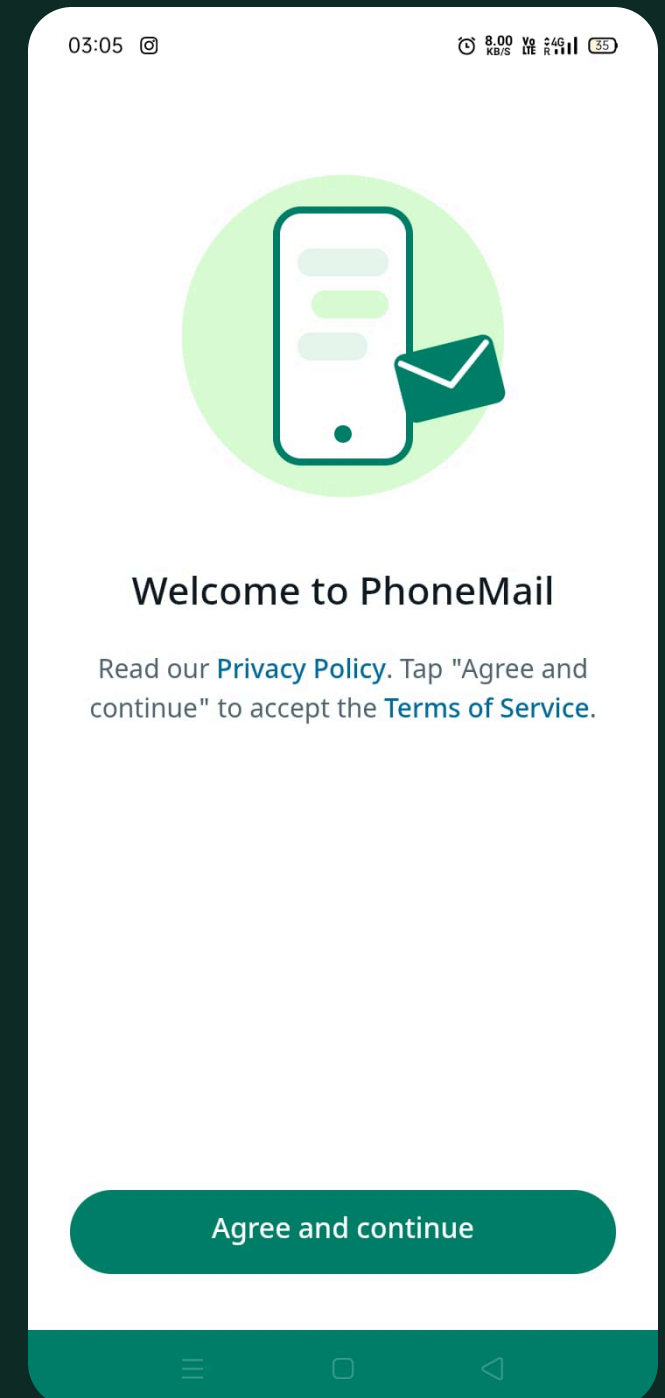
ALPHASTACK 7-DAY BUILDATHON · NIT TRICHY

# PhoneMail

Email where your phone number is your address.

9876543210@phonemail.com

Ashish Bharti



## THE IDEA

# Real email, as easy as a chat

**1**

### **Your number is your address**

9876543210@phonemail.com. It runs its own mail server, so other mail servers can deliver to it.

**2**

### **Join the way you like**

Call and press 1, text JOIN, a registration portal, or sign in with a one-time code.

**3**

### **Two faces, one mailbox**

WhatsApp-style chats on the phone, a Gmail-style inbox on the laptop, over the same mail.

**4**

### **An SMS when you need it**

"You have received an email from ... Subject: ..." for people who are not on the mobile app.

## JOINING

# Four ways in, one account behind them

**1**

### Call and press 1

A Twilio IVR reads your new address digit by digit; an SMS confirms it.

**2**

### Text JOIN

To the SMS gateway phone; PhoneMail texts back your address.

**3**

### Registration portal

/register: number and code, fields reset for the next person.

**4**

### Sign in with a code

In either app. On Android, Chrome reads the code from the SMS.

**Every path ends in the same account-creation code, with the same rules and limits.**

ON THE PHONE

# WhatsApp-style chats

## Chats grouped by people

The same people always land in the same chat; two or more make a group.

## Swipe to reply, once

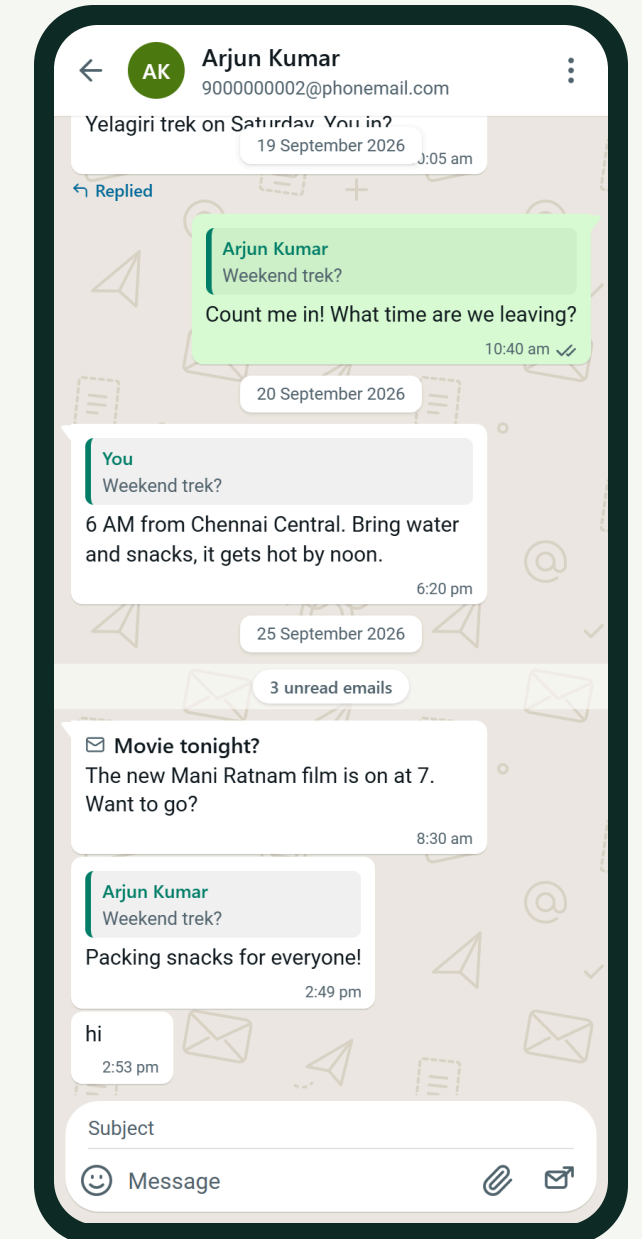
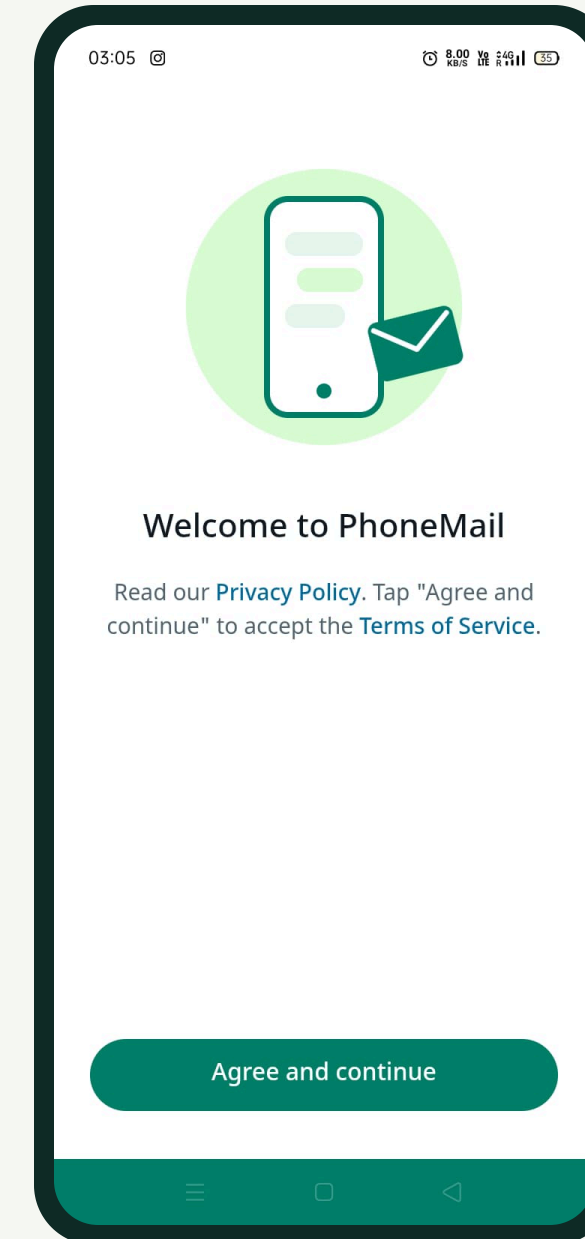
Replies are linked to the original, and each email can be answered once.

## Codes filled in for you

The sign-in code is read from the SMS, in the browser and in the Android app.

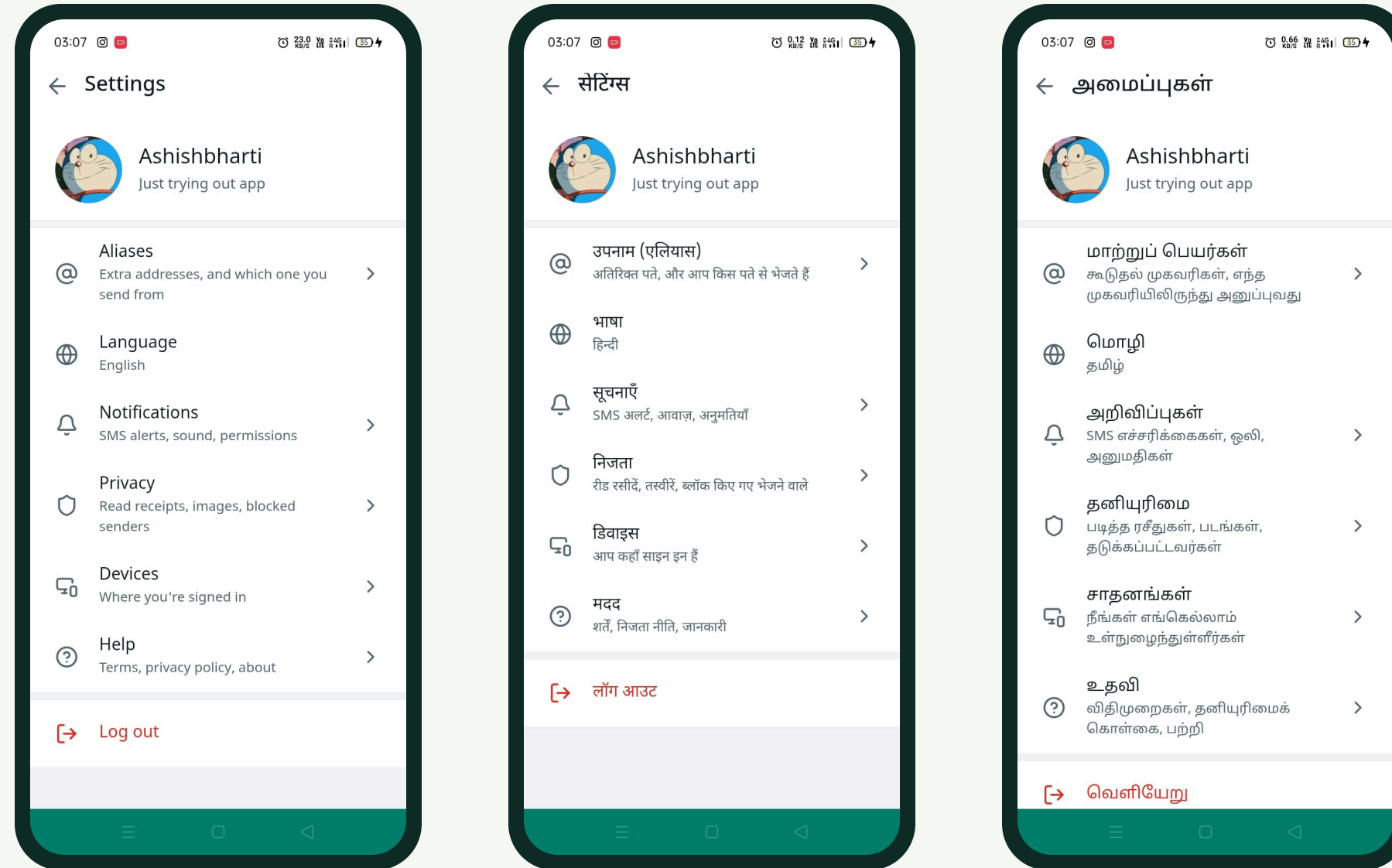
## Installs as an app

A PWA, and an Android APK with no browser bar.



FOR INDIA

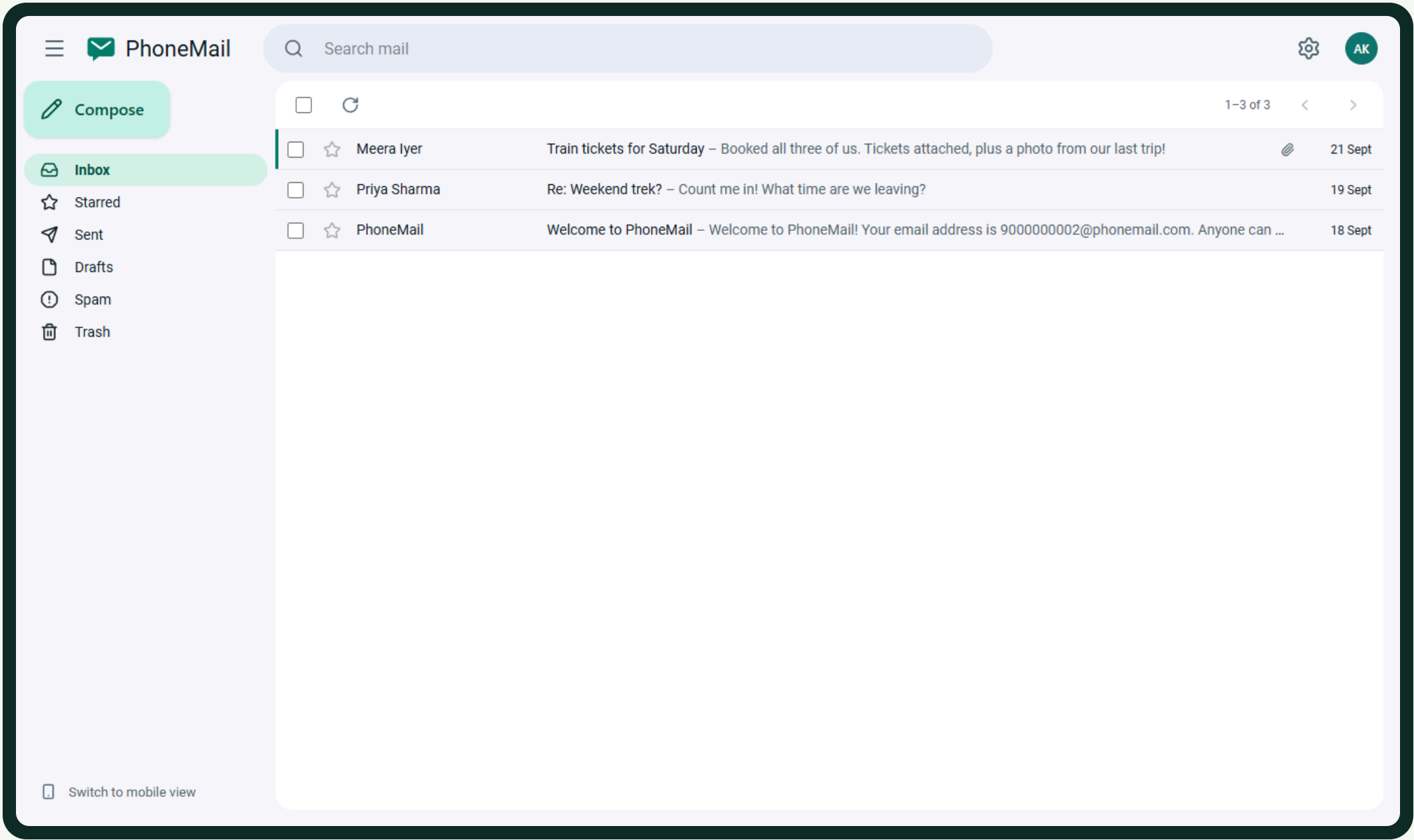
# English, Hindi and Tamil, everywhere



Every word on screen is translated; the phone call reads addresses digit by digit.

ON THE LAPTOP

# A Gmail-style inbox



## Folders and threads

Inbox, Starred, Sent, Drafts, Spam, Trash.

## Floating compose

Minimize, full screen, drafts save themselves.

## Keyboard first

Gmail shortcuts; press ? for the list.

## SMS

# A text only when it helps

### Who gets one

Only people with no active mobile-app session; the app shows mail live instead.

### What it says

"You have received an email from Arjun.  
Subject: Lunch." Cut to one SMS.

### Tested for real

Sign-in codes, alerts and JOIN replies delivered on Airtel.

### MSGate

An Android phone as the gateway: our exact text, free.



### Twilio

Free trial: a Twilio template text.



### Console

Demo mode: every SMS shown on screen.

## HOW IT IS BUILT

# Node.js and TypeScript, all in Docker

|                    |                                                           |                                           |                                                         |                                          |
|--------------------|-----------------------------------------------------------|-------------------------------------------|---------------------------------------------------------|------------------------------------------|
| <b>Clients</b>     | <b>Phone /m</b><br>React, WhatsApp-style                  | <b>Laptop /mail</b><br>React, Gmail-style | <b>Twilio • SMSGate</b><br>calls and texts              | <b>Mail servers</b><br>SMTP from outside |
| <b>Front doors</b> | <b>web</b><br>nginx: the app, /api, /socket.io, /webhooks |                                           | <b>smtp</b><br>our mail server: every email enters here |                                          |
| <b>Brains</b>      | <b>api</b><br>Fastify + zod + Socket.IO                   |                                           | <b>worker</b><br>BullMQ: SMS, outside mail, clean-up    |                                          |
| <b>Memory</b>      | <b>PostgreSQL</b><br>users, mail, chats (Prisma)          |                                           | <b>Redis</b><br>codes, limits, queues, live events      |                                          |

```
docker compose up -d # the whole stack, no .env needed
```

## RULES AND SAFETY

# Enforced by the server, not the app

### Rules

- Reply once to each email
- Recipients fixed inside a chat
- SMS only without the mobile app
- 30 emails a minute per person
- Alias names checked and held 30 days

### Safety

- Codes stored as keyed hashes, with limits
- Rotating sign-in tokens, CSRF protection
- Email HTML cleaned and sandboxed
- No open relay; signed webhooks
- Every input validated with zod

## QUALITY

# Tested by machines and on real phones

**208**

server unit tests

**22**

web unit tests

**21**

integration tests on the  
running stack

**14**

end-to-end smoke checks

### Every push

CI starts the whole stack from a fresh copy  
with no .env and runs everything.

### Real devices

Phone call, SMS alerts, JOIN sign-up, auto-  
read codes and the APK.

### Accessible

WCAG AA contrast for every colour pair,  
keyboard access, screen-reader labels.

BEYOND LOCALHOST

# Real calls, real texts, a real app

| Piece               | What it adds                                                        |
|---------------------|---------------------------------------------------------------------|
| Twilio (free trial) | A real phone call that plays the IVR menu, and template SMS alerts  |
| SMSSGate            | An Android phone as SMS gateway: exact text, codes and JOIN sign-up |
| Tailscale Funnel    | A fixed public HTTPS address for phones and webhooks                |
| Android APK         | The phone app installed full screen, with codes read automatically  |
| Mailpit             | Catches mail to outside addresses in demo mode                      |

**All optional: with no accounts at all, everything works on localhost with the demo console.**

TRY IT

# Two commands

```
git clone https://github.com/Ashish000011/Phonemail.git  
cd Phonemail && docker compose up -d  
# then open http://localhost:8080
```

## README.md

Quick start, demo story, every feature and where it lives.

## docs/

Architecture, security, 87 design decisions, the full spec.

## Walkthrough

A 90-second video on a real phone, in docs/demo.

# Thank you.